

Game 03 — Sliding Puzzle (Complete Plan)

Game 3 of 7 Complexity: ★★ Medium Estimate: 1–2 days Status: Built

Revision 2 (2026-09-11, owner-approved). The game is built and live (P0–P2 plus 5×5, picture modes, 8 procedural scenes, corner target preview, daily challenge, hard mode). This revision adds the approved **upgrade roadmap (Phases 0–5)**: a modular file structure, campaign levels + free play, a curated picture library with a builder tool, same-device + own-device multiplayer behind a config flag, and local-first persistence with mid-round resume. Supersedes the 2026-09-09 sample and revision 1.

1. Overview

Classic 15-puzzle: slide tiles into order. Hook is the personal-record pair ("4×4 in 2:41 · 213 moves"), guaranteed-solvable shuffles, and picture rounds assembled from a target image shown in the board's corner. Goal of this revision: make the game a **flagship, future-proof title** — more content (pictures, levels), more modes (daily, hard, multiplayer), and an architecture that grows without rework, while keeping the repo's constraints: vanilla ESM, no build step, no dependencies, offline-capable, games isolated from the website product.

2. Game spec (live today)

- **Sizes:** **3×3** (default), **4×4** (classic), **5×5**. Tiles $1..N^2-1$, blank last.
- **Shuffle:** from solved, `movesCount` random legal single-tile slides — 120 (3×3), 250 (4×4), 400 (5×5) — never undoing the previous move. Solvable by construction; inversion-parity (`isSolvable`) is verified in tests anyway.
- **Move counting:** single-tile slide = 1 move; row/column segment push = 1 move per displaced tile.
- **Timer:** starts on first move, stops on the winning move, accumulates across pauses (`performance.now()` only), never runs in `menu` / `won`.
- **Win:** tiles in order, blank last → overlay. Input locked after the winning move.
- **Score:** `max(0, sizeBonus - ⌊moves/2⌋ - ⌊seconds/5⌋)`, sizeBonus 300/500/800.
- **Wrong-tile feedback:** shake animation, no move counted.
- **Modes (live):** Numbers or Picture tiles; **Daily challenge** (seeded 4×4 — same board for everyone that day, per-day record); **Hard** (picture-only, no preview/peek, separate records).

3. Progression model (this revision)

- **Campaign:** sequential stages with difficulty ramping (grid size, shuffle depth, fading assists, par moves/time). A stage unlocks when the previous is solved. Stars (1–3) from par thresholds; replays improve stars. Per-stage best time/moves.

- **Free Play:** today's menu verbatim (any size, numbers/picture, daily, hard) — always available, **no unlocks, no stars, no campaign progress** (its limitation). Per-size/mode records still apply.
- **Daily:** unchanged — seeded 4×4, per-day record; picture becomes deterministic from the same seed (planned, Phase 1) so every player sees the identical board _and_ image.
- **Records:** per size/mode/variant (`best:<size>[:hard]`), per day (`daily:<yyyymmdd>`), and per campaign stage (`progress`), all localStorage, schema-versioned.

4. Screens & UI (this revision)

Screen	Elements
Menu (hub)	Section entry points: Campaign, Free Play, Multiplayer ; mute; theme follows site
Campaign	Stage grid: lock state, stars, best time/moves per stage; continue point
Free Play	Today's menu: size picker (3 cards with bests), numbers/picture, hard, daily (with date + done state)
Playing	Compact centered cluster: HUD (menu · timer · moves · pause), board (<code>--n</code> -driven), corner target preview (picture rounds), actions (peek · restart · shuffle)
Paused	Opaque overlay — board hidden, clock held; resume/restart/menu
Won	Time, moves, score; NEW BEST badges; campaign variant adds stars + [Retry] [Next stage]
Multiplayer	Series (same device) and challenge link (own device); entry gated by flag (§7)

Tiles are absolutely-positioned buttons moved via `transform: translate` transitions (180 ms ease); the blank is a missing tile. Board sizing `min(92vw, 60dvh)`, capped for desktop; HUD + board + actions cluster together at every viewport.

5. Data model (localStorage, schema-versioned, guarded)

```
game:sliding-puzzle:prefs      { version, size, muted, mode, hard, locale? }
game:sliding-puzzle:best:<size>[:hard]  { timeMs, moves }
game:sliding-puzzle:daily:<yyyymmdd>    { timeMs, moves }
game:sliding-puzzle:progress      { version, unlockedUpTo, stages:{ id:{timeMs,moves,stars} } }
game:sliding-puzzle:resume      { version, size, seed?, board, moves, playedMs, mode, assists, stageId? }
```

All reads validate + migrate by `version`; all writes best-effort (storage disabled → game fully playable). A pluggable **remote adapter slot** exists in `storage.js` for a future host-injected account sync — the game never checks auth and makes no network calls (§7).

6. File structure (modular — replaces the original 4-file layout)

```
sliding-puzzle/
  index.html      screens: menu hub | campaign | free play | playing | multiplayer
  style.css       tokens (light/dark), screens, board, overlays, reduced-motion
  core.js         pure model: board, shuffle (+guards), slideTile (+displaced tiles),
                  solvability, scoring, clock format, per-size config, key builders,
                  mulberry32, mergeRecord — no DOM (the test surface)
  levels.js       [P1] stage definitions + unlock/star rules — pure
  scenes.js       [P0] procedural painters + makePicture + sliceBackground
  library.js      [P2] picture-library manifest access — pure
  config.js       [P0] flags + strings, host-overridable (§7)
  storage.js      [P0] guarded storage facade: version + migrations, prefs, records,
                  progress, resume, remote adapter slot
  audio.js        [P0] WebAudio blips + mute (context on first gesture)
  game.js         UI shell: screens, input, timer, share (imports the modules above)
  data/pictures.json [P2] library manifest (empty until packs are added)
  pics/           [P2] bundled square images (only when the owner adds them)
  builder.html    [P2] Library Builder tool (local, not linked from the game)
  core.test.html  in-browser harness for the pure modules (auto-runs)
```

7. Multiplayer & the website/app flag

Modes (both backend-free):

- **Same-device series:** two players alternate on identical seeded boards; per-round result + series scoreboard (Tic Tac Toe's pass-and-play pattern). Separate records.
- **Own-device challenge:** a challenge encodes `{size, seed, assists, pictureId}` in a URL/code; the friend plays the `_identical_` board on their device; results compared via share text + a manual "opponent result" entry. Nothing at stake → no trust problem.

Website/app flag (config.js):

```
multiplayerEnabled: false // website default; the app wrapper injects true
```

Resolution order: host-injected `window.__SLIDING_PUZZLE_CONFIG__` → `?mp=0|1` URL param

→ default. One file, no refactor to switch environments (the flag arrives when that static file is re-served/injected — cache-busting in §12).

- `true` → multiplayer behaves normally.
- `false` → the Multiplayer entry stays **visible but disabled**: click opens a small dialog with the configurable notice ("Multiplayer available in the app only."), rendered `aria-disabled`; **no multiplayer logic executes, no network calls**. Modules stay loaded but inert. Notice text lives in `config.strings` (per-locale); `prefs` may override it but is a cache, not the source of truth.

Gated — explicitly NOT built (needs backend + \$7 reversal): live rooms, real-time races, in-app leaderboards, account-level sync. Client `performance.now()` cannot be trusted competitively; any of these requires server-side timing and an explicit, separately-scoped owner change reversing the isolation decision.

8. Persistence (local-first)

- **Guests and the website build: full local persistence** — records, daily, campaign progress, resume. Dropping guest saving would erase most players' progress and break the offline contract; not done.
- **Resume:** an unfinished round snapshots `{size, seed?, board, moves, playedMs, mode, assists, stageId?}` (~200 B) and offers "Continue" on return — for everyone, guests included.
- **Logged-in continuity is a host concern:** `storage.js` exposes a remote adapter slot; the website/app injects an adapter for authenticated users (local-first, best-effort mirroring). The game never checks auth → no coupling, no network in the website build.
- **Account-level/cross-device sync — gated future phase (\$7):** needs the backend and an explicit isolation reversal. Documented, not built.

9. Testing

- **core jest suite** (`src/__tests__/games-sliding-puzzle.test.ts`): model proofs — 100 shuffles/size solvable + never solved + no-undo replay through `slideTile`; segment slides; solvability formulas (odd/even width); scoring; seeded daily determinism; record merging; plus suites for the new pure modules: per-size config resolution, shuffle guards, levels (unlock/star boundaries), storage migrations, library manifest parsing/fallback, config flag resolution, `sliceBackground`.
- **core.test.html** harness mirrors the pure assertions in-browser (auto-runs).
- **UI layer** (`game.js`) is verified by browser passes (flag states, layouts at 360 px and desktop, resume accuracy ± 50 ms) — UI logic stays thin by pushing rules into the pure modules above.
- CI safety: every committed test imports only tracked files.

10. Build phases (approved roadmap)

Phase 0 — Foundation: correctness fixes, config/flags, storage facade

- ☐ `shuffle()` guards: validate `movesCount` (unknown size → clear error), bound the solved-retry loop (no infinite loop from a degenerate rng); pathological case falls back to a forced nudge.
- ☐ One per-size config object (`shuffleMoves`, `sizeBonus`) as the single source; `SIZES / SHUFFLE_MOVES / SIZE_BONUS` derived from it; unknown size fails fast.
- ☒ `slideTile` returns displaced tile values; UI stops re-implementing the geometry. (2026-09-12: returns `{ board, moved, pushed }`; `pushedTileValues` deleted.)
- ☒ `config.js` + `storage.js` + `scenes.js` + `audio.js` modules created; `game.js` becomes the UI shell importing them (behavior unchanged). (2026-09-12)
- ☒ Dead CSS removed (`board--deal`, `overlay--in`, `@keyframes deal/pop`); `.daily` added to reduced-motion. (2026-09-12)
- ☐ Score/clock rounding unified.

- ☒ Jest + harness updated for the new modules; all green; game behavior unchanged. (2026-09-12: versioned `game:sliding-puzzle:save` v1 with legacy-key migration.)

Phase 1 — Campaign levels + Free Play

- ☐ `levels.js`: stage definitions (~24 stages, three tiers: 3×3 → 4×4 → 5×5, assists fading, fixed picture per stage, par + star thresholds) and pure rules (`stageById`, `isUnlocked`, `starsFor`, `nextStage`).
- ☐ `storage.js`: progress record (versioned, migrated) + free-play separation.
- ☐ UI: Campaign screen (grid, locks, stars, bests), Free Play screen (today's menu), campaign win overlay (stars, retry/next), hub navigation.
- ☐ Daily picture deterministic from the daily seed.
- ☐ Jest: unlock/star/scoring boundaries; browser: unlock flow + free-play parity.

Phase 2 — Picture library + Library Builder

- ☐ `data/pictures.json` manifest schema (packs, pictures, sections) + `library.js` (resolve/fallback) + `scenes.js` picture resolution for levels/daily/free.
- ☐ `builder.html`: upload → square-crop/resize 720×720 → JPEG q0.85 → assign packs + sections → export `pictures.json` + named `pics/<pack>/<id>.jpg` files.
- ☐ Rules: ≤ ~150 KB per image, square, license-clear (owner's responsibility); zero entries → procedural-only behavior unchanged.
- ☐ Jest: manifest validation/resolution/fallback; browser: assigned image slices correctly + appears in the corner preview; builder round-trip.

Phase 3 — Multiplayer + website/app gating

- ☐ Same-device series: alternating seeded boards, per-round results, series scoreboard.
- ☐ Own-device challenge: URL/code encoding `{size, seed, assists, pictureId}`; identical board + picture on the friend's device; result comparison via share + manual opponent entry.
- ☐ Flag gating per §7: disabled entry + dialog + zero logic/network; `?mp=1` /host override enables full behavior.
- ☐ Jest: challenge encode/decode round-trip; browser: both flag states, network-request log empty when disabled; isolation grep clean.

Phase 4 — Persistence & resume

- ☐ `storage.js` migrations + remote adapter slot (host-injected only).
- ☐ Mid-round resume snapshot (save on state change, offer Continue on return); works for guests; campaign rounds resume into their stage.
- ☐ Docs: host-injection contract for the adapter (the game never checks auth).
- ☐ Jest: snapshot save/restore round-trip, migration path; browser: resume accuracy ±50 ms; zero network calls with no adapter injected.

Phase 5 — Polish, accessibility, QA, docs

- ☐ A11y: tile position in `aria-label` + blank announcement (polite live region), focus trap/inert for `aria-modal` dialogs, roving focus + arrow keys for radio groups, focus target when a round starts.

- ☐ Contrast to AA (primary gradient light stop, muted text); `color-mix` fallback + `-webkit-backdrop-filter`.
- ☐ Cache-busting for `style.css` / `game.js` / `data/pictures.json`.
- ☐ Timer/share helper extraction + tests; determinism test fixed (two fresh streams); mid-board segment + non-integer index cases.
- ☐ Docs: this file's status + `plan/games/README.md` row per phase; gated items (§7) recorded.
- ☐ QA gate (master README §5): offline check, isolation greps, 10-minute phone session (incl. iOS Safari fallbacks).

11. Acceptance criteria

- All Phase checklists above checked; jest + `core.test.html` green; CI safe (tests import only tracked files).
- Shuffle can never hang regardless of config or rng; unknown sizes fail fast with a clear message.
- `multiplayerEnabled:false` (website default) = single-player only, zero network calls, verified in the browser's request log.
- Timer exact across pause/resume (± 50 ms); resume restores board/moves/clock exactly.
- Campaign unlock/star rules correct at boundaries; Free Play never awards progression.
- Library: zero entries → today's behavior; assigned images slice correctly and appear in the corner preview; builder round-trips assignments.
- 3×3 fits 360 px with HUD visible, no scroll; layouts compact at phone and desktop.
- Restart ≤ 1 tap; shuffle visibly mixes the board every time.

12. Deferred coupling & gated items (intentionally NOT built)

- Footer/nav links, analytics events, achievements integration, backend leaderboards — per master README §7. Sliding Puzzle adds **no** analytics and **no** site coupling.
- **Gated future phases (require backend + explicit §7 reversal):** live rooms/real-time races, in-app leaderboards, account-level cross-device sync. Server-side timing validation is mandatory for anything competitive.
- Shipping the games publicly (removing `.gitignore` isolation lines + footer/sitemap/ Play Hub references) remains a separate, explicitly-scoped owner action. Sliding Puzzle is the only committed game (owner decision 2026-09-11).